Early Black-Hole Growth Constraints from JWST

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Abstract

Aims. Massive black holes in the early Universe must grow within limited cosmic time. We identify which JWST black-hole candidates most strongly constrain early growth models and which assumptions control these constraints. **Methods.** We compile 350 literature measurement records from 32 source families at $z \geq 4$. We analyse 224 objects with broad hydrogen-line mass estimates and include 10 additional objects in a broader comparison. We propagate reported mass uncertainties and test mass-scale shifts and alternative growth assumptions. **Results.** For $100\,M_\odot$ seeds formed at $z = 30$, radiative efficiency $\epsilon = 0.1$, and no merger boost, 12 of the 224 objects require a lifetime-average Eddington ratio $\overline{f_{\mathrm{Edd}_\mathrm{req}}} > 1$. Of these, 6 exceed this threshold in at least 95% of trials sampling their quoted mass errors under fixed growth assumptions. UNCOVER-20466, COSMOS3D-13852 and RUBIES-EGS-55604 retain this criterion after a -0.5 dex mass shift, although estimator systematics remain. Delaying seeding to $z = 20$ makes GN-z11 the most demanding object. For A2744-QSO1, an independent dynamical mass estimate raises its requirement to approximately unity, illustrating how mass measurements affect growth constraints. **Conclusions.** At $\epsilon = 0.05719$, the ideal thin-disc efficiency for a non-spinning black hole, all 234 central mass estimates are reachable from $100\,M_\odot$ seeds formed at $z = 30$ without lifetime-average accretion above the Eddington limit. We distinguish seed and efficiency assumptions that permit Eddington-limited average growth within the available time from those requiring higher rates, and identify the strongest observational constraints.

Key words. black hole physics – accretion, accretion disks – galaxies: active – galaxies: high-redshift – quasars: supermassive black holes

1 Introduction

The discovery of massive black holes in the early Universe raises a fundamental question: what growth histories could have assembled them within the available cosmic time? The strength of this constraint depends on both the observed black-hole mass and redshift, as well as assumptions about seed formation, radiative efficiency, and the duration of active accretion. An apparently demanding object under one set of assumptions may therefore be compatible with less extreme growth under another.

Here, we use a literature-based catalogue of JWST-identified high-redshift accreting black holes and candidates to test which combinations of seed mass, formation time, radiative efficiency and accretion rate can reproduce the observed masses by the observed redshifts. The atlas covers $z \geq 4$, extending from the first billion years to a cosmic age of approximately 1.54 Gyr at its lower-redshift boundary. We calculate the average accretion rates required across a range of seed and growth assumptions, incorporating reported mass uncertainties and distinguishing measurements with different evidential and methodological limitations. Source-native measurements, identity links, and mass provenance are retained to support comparison across heterogeneous literature samples.

Our aim is to identify which growth requirements remain demanding across plausible scenarios, which depend primarily on uncertain assumptions, and where improved observations would most strengthen the constraints. This approach treats the catalogue as a means of testing possible growth histories, rather than assuming that every high-redshift black hole presents the same assembly problem. The constraints apply to individual objects under stated assumptions.

Analytic seed-growth comparisons already establish the trade-off between seed mass and sustained accretion. For example, P. Dayal (2024) considered light and heavy seeds at $z_{\rm seed} = 25$ and explored a primordial-seeding interpretation. Here we use this established growth framework to assess how object-level constraints change when published masses, measurement methods, and growth assumptions are treated consistently across a larger evidence atlas. The added scientific product is an object-by-object distinction between requirements that survive quoted errors, those that survive a mass-scale perturbation, and those whose interpretation depends on a specific estimator. We report the complete primary threshold set with source-linked caveats and proposed observations, isolate measurement and starting-time effects in a matched literature comparison, and evaluate an independent dynamical estimate outside the frozen catalogue. These comparisons make the catalogue useful for selecting observational tests of growth assumptions. Survey studies provide a complementary perspective: the JADES broad-line census (I. Juodžbalis et al. 2026a) measures current accretion and host scaling relations, while the narrow-line census (J. Scholtz et al. 2025) broadens the evidence for obscured activity. Our comparison asks what cumulative growth the adopted masses require, which is distinct from measuring present accretion or counting AGN in a survey. It uses those source measurements without imposing a common selection function. The source-family inventory is given in Appendix A.

1.1 Theoretical framework

We use the luminosity-based Eddington ratio $f_{\rm Edd} = L_{\rm bol}/L_{\rm Edd}$ and a fixed radiative efficiency ϵ, the fraction of accreted rest-mass energy emitted as radiation. For the available growth interval,

$$\Delta t = t(z_{\rm obs}) - t(z_{\rm seed}) > 0, \qquad \overline{f_{\rm Edd}} = \frac{1}{\Delta t}\int_{t(z_{\rm seed})}^{t(z_{\rm obs})} f_{\rm Edd}(t)\,dt. \tag{1}$$

The analytic growth relation adopted by P. Dayal (2024) gives

$$M_{\rm BH}(t_{\rm obs}) = B_{\rm merge} M_{\rm seed} \exp\left[\frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}}\right], \tag{2}$$

where $t_{\rm Edd} \simeq 0.45$ Gyr. The merger multiplier is $B_{\rm merge} = 1$ for the reference calculation and 2 for a factor-of-two mass boost. This prescription rescales the seed normalization; it does not simulate merger times, mass ratios, gravitational-wave losses or a merger tree. The required lifetime-average ratio is

$$\overline{f_{\rm Edd}}_{\rm req} = \max\left[0, \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{\Delta t}\ln\left(\frac{M_{\rm BH}}{B_{\rm merge}M_{\rm seed}}\right)\right]. \tag{3}$$

If the boosted seed already exceeds the observed mass, the requirement is set to zero; this does not make the specified history an exact mass match. A lifetime-average requirement is distinct from an instantaneous Eddington ratio and does not uniquely determine a duty cycle.

Cosmic ages assume a flat matter-plus-Lambda cosmology with $H_0 = 67.3\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$ and $\Omega_\Lambda = 0.685$, neglecting radiation. Appendix D gives the supporting relations, implementation and inverse seed-mass diagnostic. The fixed-efficiency assumption is essential to Eqs. 2–3. The supplementary spin-dependent compatibility prescription is specified in Appendix C; it does not model spin evolution.

2 Catalogue construction

We compile 350 literature measurements from 32 source families at $z \geq 4$. Each adopted mass is linked to its source, method and reported uncertainty. One preferred measurement is selected per object using documented source assessments; alternate measurements remain available for sensitivity tests and are not averaged into the preferred value.

Growth analysis requires a numerical mass and redshift, an identified mass method and a resolved lensing treatment. Upper limits, indirect model ranges and masses inferred by assuming an Eddington ratio are excluded from numerical inference. We also exclude all six records associated with three unresolved identity groups, including three records with eligible masses. This leaves 224 primary objects and an inclusive exploratory sample of 234 objects.

The primary sample uses Balmer-line single-epoch virial masses with secure or probable evidence and no conditional broad-line-region interpretation. The ten additional exploratory objects comprise nine candidates and GN-z11, whose UV mass estimator lies outside the primary method group. Missing mass errors do not exclude a point estimate, but prevent probability inference. These cuts improve method comparability without implying a common calibration or survey selection function. Appendix A gives the source inventory, identity decisions and sample accounting (Figure 6).

3 Growth-model comparison

3.1 Reference model

We calculate the required lifetime-average Eddington ratio using Eq. 3, with $M_{\rm seed} = 10^2 M_\odot$, $z_{\rm seed} = 30$, $\epsilon = 0.1$ and $B_{\rm merge} = 1$ as the reference history. We refer to this conditional requirement as *growth pressure*. Figure 1 illustrates tracks at four fixed efficiencies; tracks are selected for proximity to the observed masses, not fitted growth histories. Appendix D.6 specifies the display selection.

3.2 Uncertainty and sensitivity

We propagate reported asymmetric log-mass errors through 10,000 draws per object, assigning half the draws to each side of the central mass with the corresponding half-normal scale. Redshifts are held fixed. This approximates quoted intervals rather than reconstructing source posteriors. The 12 primary objects without reported errors contribute point estimates only; probability calculations use 212 primary and 222 inclusive exploratory objects. No additional calibration scatter is added to the quoted errors.

We test mass-scale sensitivity by shifting all adopted log masses and their error draws by offsets of -0.5, -0.3, 0, $+0.3$ and $+0.5$ dex, keeping the sample and growth assumptions fixed. These offsets probe the single-epoch mass uncertainty scale (Y. Shen 2013); they are stress tests, not inferred calibration corrections. Probabilities remain conditional on each offset. Sampling and conversion details are given in Appendix D.6.

3.3 Compatibility across growth assumptions

A fixed history is compatible with an object when its required seed mass (Eq. 15) lies within the chosen interval in $\log_{10}(M_{\rm seed}/M_\odot)$: $[1, 2]$ (light), $[3, 4]$ (intermediate), $[4, 6]$ (heavy), or $[2, 6]$ (broad). The grid uses $z_{\rm seed} = 30$, three spin/efficiency cases, merger multipliers 1 and 2, and average rates 0.3, 1 and 2. Unlike the reference ranking and fixed-efficiency growth tracks, these maps use the supercritical efficiency prescription in Appendix C.

Seed intervals overlap and carry no probability weights; the broad interval has no formation-channel interpretation. A required seed below an interval means that the specified history overgrows the observed mass. Compatibility fractions describe the admitted objects, not population frequencies, because the literature sample has no common selection function.

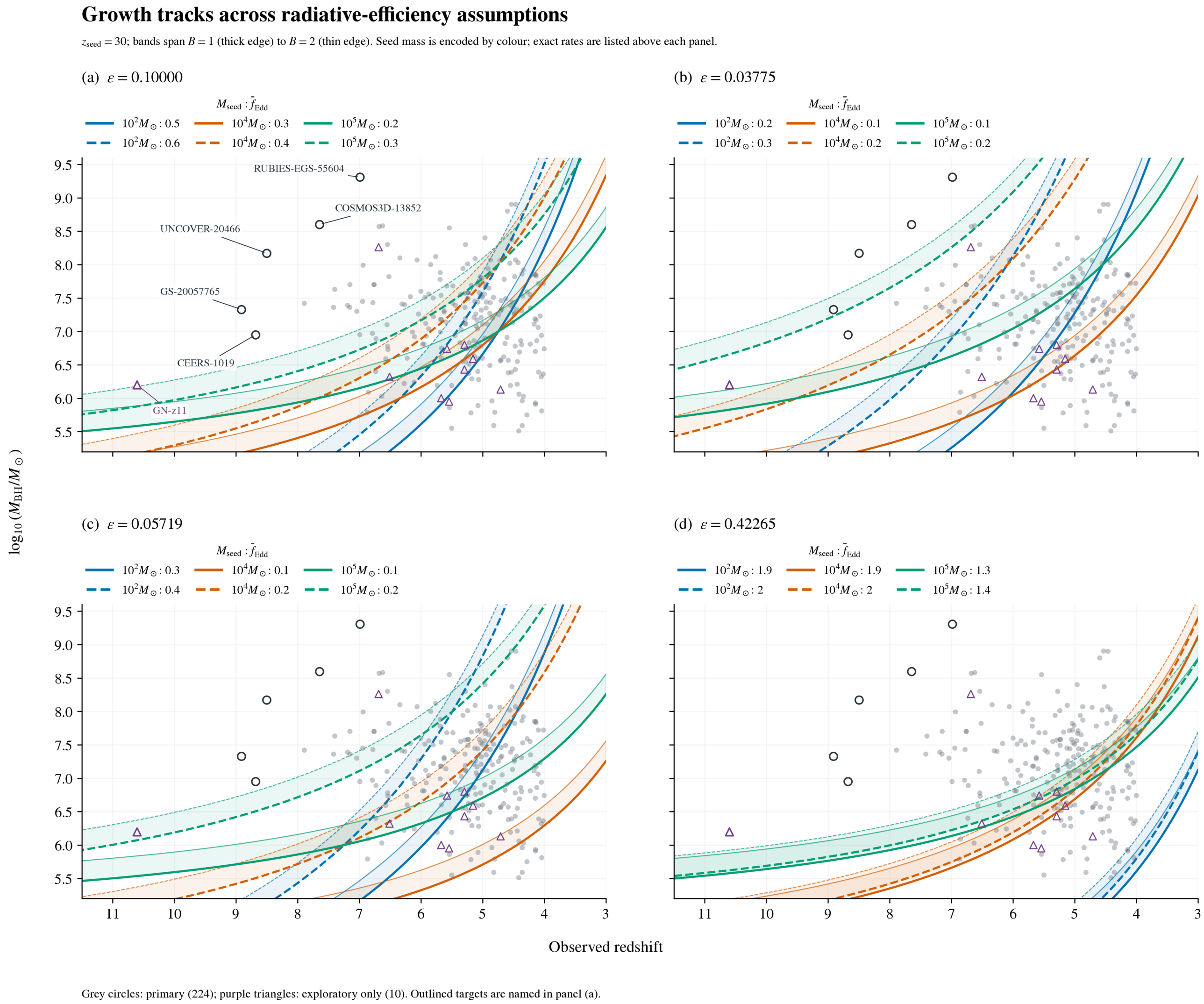

Figure 1: Growth tracks with 224 primary and ten exploratory-only objects. Panels show four constant radiative efficiencies at $z_{\rm seed} = 30$; colours distinguish seeds of 10^2, 10^4 and $10^5 M_\odot$, and legends list the selected average Eddington ratios. Shaded bands span merger boosts $B = 1$–2, with thick and thin edges respectively; they are not uncertainty intervals. Six named objects are outlined in every panel. The reversed redshift axis spans $z = 11.5$ to $z = 3$. Excluded identities and objects without eligible masses are omitted.

4 Results

4.1 Growth pressure under the reference assumptions

For a $10^2\ M_\odot$ seed under the reference history, the 224-object primary sample has median required $f_{\rm Edd} = 0.578$ (interquartile range 0.487–0.682), with 12 point requirements above unity and none above two. The inclusive 234-object sample has median 0.574 (0.488–0.679) and 14 above unity. UNCOVER-20466 is most demanding: its point requirement is 1.452 and its reported-error median is 1.453 (16th–84th percentiles 1.354–1.549). Table 1 lists the 12 primary threshold objects; their ordering by point requirement differs from ordering by quoted-error probability.

Growth pressure depends on time as well as mass: GN-z11 ranks second at $z = 10.603$ despite its smaller mass. Its UV virial estimator places it in the exploratory sample.

Table 1: Complete primary reference-threshold set, ordered by point requirement. Masses and errors are in $\log_{10}(M/M_{\odot})$; all estimators are single-epoch virial. f_0 is the reference point requirement, f_{16} its reported-error 16th percentile, and $P_0 = P(\overline{f_{\mathrm{Edd}_{\mathrm{req}}}} > 1)$. f_- and P_- apply the fixed -0.5 dex offset. Rounded probabilities above 0.999 do not imply physical certainty. Source-specific caveats and proposed observations are in Table 6.

Object/source	Line	Log mass	f_0	f_{16}	P_0	f_-	P_-
UNCOVER-20466	Hβ	$8.17^{+0.42}_{-0.42}$	1.452	1.354	> 0.999	1.335	> 0.999
J. E. Greene et al. (2024)							
GS-20057765	Hβ	$7.33^{+0.62}_{-0.70}$	1.355	1.177	0.980	1.228	0.901
I. Juodžbalis et al. (2026a)							
COSMOS3D-13852	Hβ	$8.60^{+0.35}_{-0.33}$	1.314	1.247	> 0.999	1.214	> 0.999
X. Lin et al. (2026)							
RUBIES-EGS-55604	Hβ	$9.31^{+0.10}_{-0.10}$	1.267	1.250	> 0.999	1.180	> 0.999
D. D. Kocevski et al. (2025)							
CEERS-1019	Hβ	$6.95^{+0.37}_{-0.37}$	1.205	1.115	0.988	1.083	0.822
R. L. Larson et al. (2023)							
GS-20030333	Hβ	$7.42^{+0.65}_{-0.48}$	1.133	1.033	0.907	1.029	0.613
I. Juodžbalis et al. (2026a)							
GS-164055	Hβ	$7.63^{+0.59}_{-0.66}$	1.065	0.941	0.696	0.970	0.395
I. Juodžbalis et al. (2026a)							
GN-38509	Hα	$8.57^{+0.37}_{-0.38}$	1.065	1.005	0.860	0.984	0.392
I. Juodžbalis et al. (2026a)							
J0910_2028_12910	Hα	$8.58^{+0.01}_{-0.01}$	1.053	1.051	> 0.999	0.973	< 0.001
C. J. Baccus & X. Xu (2026)							
CEERS-00717	Hα	$7.99^{+0.16}_{-0.17}$	1.027	0.998	0.821	0.942	0.018
Y. Harikane et al. (2023)							
ZS7	Hβ	$7.70^{+0.40}_{-0.40}$	1.023	0.954	0.625	0.934	0.173
H. Übler et al. (2024)							
GN-4685	Hβ	$7.36^{+0.45}_{-0.42}$	1.017	0.938	0.585	0.922	0.181
I. Juodžbalis et al. (2026a)							

4.2 Robustness to reported mass uncertainty

Of the 12 primary point requirements above unity, eight remain above it at the 16th percentile and six have conditional $P(f_{\mathrm{Edd}} > 1) \geq 0.95$ (Fig. 2). The exploratory counts are 14 at the median, ten at the 16th percentile and eight at $P \geq 0.95$; 15 exceed unity by the 84th percentile. Probabilities follow Section 3.2 and use 212 primary and 222 exploratory objects with reported errors. The 12 primary objects without errors enter only point counts.

4.3 Sensitivity to the virial-mass scale

Downward mass shifts of 0.3 and 0.5 dex reduce the primary point/16th-percentile/ $P \geq 0.95$ counts from 12/8/6 to 9/6/4 and 6/4/3, respectively (Table 2). UNCOVER-20466, COSMOS3D-13852 and RUBIES-EGS-55604 retain $P \geq 0.95$ at -0.5 dex. They warrant estimator tests, but this common offset neither brackets every source-specific bias (Section 5.2) nor removes the growth-model degeneracies.

Replacing the redshifts, masses and errors of 44 exact-ID, position-confirmed Baccus counterparts with published Table 1 values (C. J. Baccus & X. Xu 2026) preserves the primary 12/8/6 and exploratory 14/10/8 counts and both top-five rankings (Table 3). The 32 changed central masses shift by -0.10 to $+0.04$ dex. Of five unmatched records, one is identity-excluded; retaining or omitting the other four leaves these results unchanged (220/230 objects when omitted). The test holds all six identity exclusions, method classes, growth parameters and sampling rules fixed; it does not admit the full published sample

Objects above the reference growth threshold

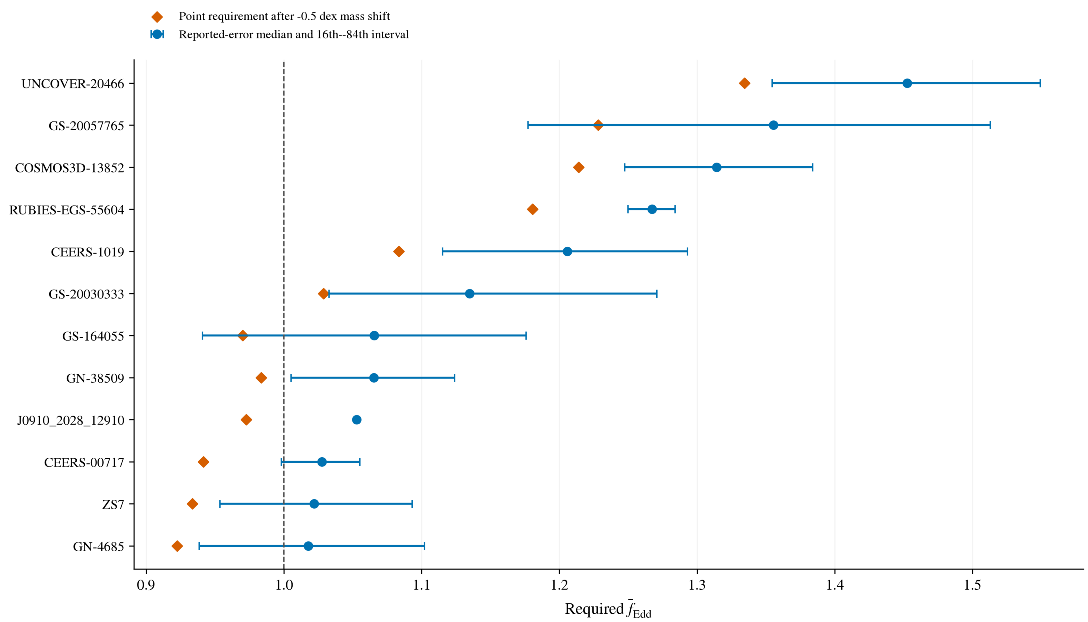

Figure 2: The 12 primary objects whose reference point requirements exceed unity. Circles and bars show reported-error medians and 16th–84th percentile intervals; diamonds show point requirements after a -0.5 dex mass shift. The vertical line is the reference threshold. The selected objects all have reported errors; these intervals do not include additional estimator systematics.

Table 2: Fixed mass-scale stress tests under the reference growth history. Triplets give counts above unity at the point estimate, at the 16th percentile, and with conditional $P(\overline{f_{\rm Edd}}_{\rm req} > 1) \geq 0.95$. The probability-based counts use 212 primary and 222 exploratory objects with reported errors; point counts use all 224 and 234 objects.

Mass offset (dex)	Primary	Exploratory
-0.5	6 / 4 / 3	7 / 5 / 4
-0.3	9 / 6 / 4	10 / 7 / 5
0	12 / 8 / 6	14 / 10 / 8
+0.3	15 / 13 / 12	17 / 15 / 14
+0.5	19 / 16 / 13	21 / 18 / 15

or resolve the excluded identities.

4.4 Model compatibility and measurement choice

Increasing the seed mass to $10^3\,M_\odot$ reduces the primary count above unity from 12 to four; $10^5\,M_\odot$ seeds remove all point exceedances. Delaying seeding to $z = 20$ raises the primary count to 22, while $\epsilon = 0.05719$ removes all point exceedances in both samples (Table 4). These one-parameter tests span light/heavy seeds (P. Dayal 2024), shorter growth time and the ideal non-spinning thin-disc efficiency (Eq. 7); they have no inferred population weights.

Removing the six ambiguous records from the original 227/237-object sets preserves threshold counts and top-five rankings in every listed scenario, including the reference probability counts. The primary median

Table 3: Joint Baccus publication-update and identity-exclusion test. Every row excludes all open identity groups. Counts denote point requirements above unity, 16th percentiles above unity, and conditional $P \geq 0.95$, respectively. Frozen values define the manuscript baseline.

Sample	Measurement treatment	Objects	Counts
Primary	Frozen values	224	12/8/6
Primary	Published; retain unmatched	224	12/8/6
Primary	Published; omit unmatched	220	12/8/6
Exploratory	Frozen values	234	14/10/8
Exploratory	Published; retain unmatched	234	14/10/8
Exploratory	Published; omit unmatched	230	14/10/8

Table 4: Point-estimate requirements above $\overline{f_{\rm Edd}} = 1$ under fixed histories. Each row changes only the named parameter from the reference ($M_{\rm seed} = 10^2 M_\odot$, $z_{\rm seed} = 30$, $\epsilon = 0.1$, $B_{\rm merge} = 1$). Counts use provisional object identities and do not include uncertainty tails.

Scenario	Primary (224)	Exploratory (234)
Reference	12	14
$M_{\rm seed} = 10^3 M_\odot$	4	5
$M_{\rm seed} = 10^5 M_\odot$	0	0
$z_{\rm seed} = 20$	22	24
$\epsilon = 1 - \sqrt{8/9}$	0	0

changes from 0.574 to 0.578; the early-growth tail is unaffected by these exclusions.

Figure 3 shows the maximum fixed efficiency allowing $\overline{f_{\rm Edd}}_{\rm req} \leq 1$ for five targets as a function of seed mass. The $z_{\rm seed} = 20$ and 30 curves expose GN-z11's greater timing sensitivity. These point-mass boundaries assume no merger boost and do not establish whether the required fuel supply is sustainable.

The primary and exploratory samples show similar compatibility patterns (Fig. 4). For heavy seeds at $a = +1$ and $f = 1$, changing $B_{\rm merge}$ from one to two raises compatibility from 57 to 70 per cent in the primary sample (59 to 71 per cent inclusive). Faster growth can also overproduce the observed mass, moving the required seed below its interval. These point-mass fractions use the efficiency prescription in Appendix C; they neither integrate mass errors nor measure formation-channel frequencies.

Seven alternate-measurement comparisons across six objects (two for CEERS-2782) span mass shifts of -0.600 to $+0.949$ dex and required-$f_{\rm Edd}$ shifts of -0.083 to $+0.121$ (Fig. 5). Both estimates remain below unity in every pair. This stability applies to the six tested objects, not the untested high-pressure tail.

4.5 Separating measurement and starting-time changes

For four unambiguous matches to Table 1 of P. Dayal (2024), Table 5 first replaces the earlier masses and redshifts with our adopted values at $z_{\rm seed} = 25$, then changes seeding to $z = 30$. All columns use our cosmology, $\epsilon = 0.1$, $10^2 M_\odot$ seeds and no merger boost. Earlier seeding lowers the requirements more than the measurement updates change them. This isolates input and timing effects rather than reproducing the earlier analysis or providing a complete crossmatch.

4.6 External dynamical-mass check: A2744-QSO1

For A2744-QSO1, we compare the frozen virial mass $\log_{10}(M/M_\odot) = 7.3 \pm 0.2$ (J. E. Greene et al. 2024) with the external MOKA3D point-mass estimate 7.7 ± 0.3 (I. Juodžbalis et al. 2026b), using $z = 7.04$ and

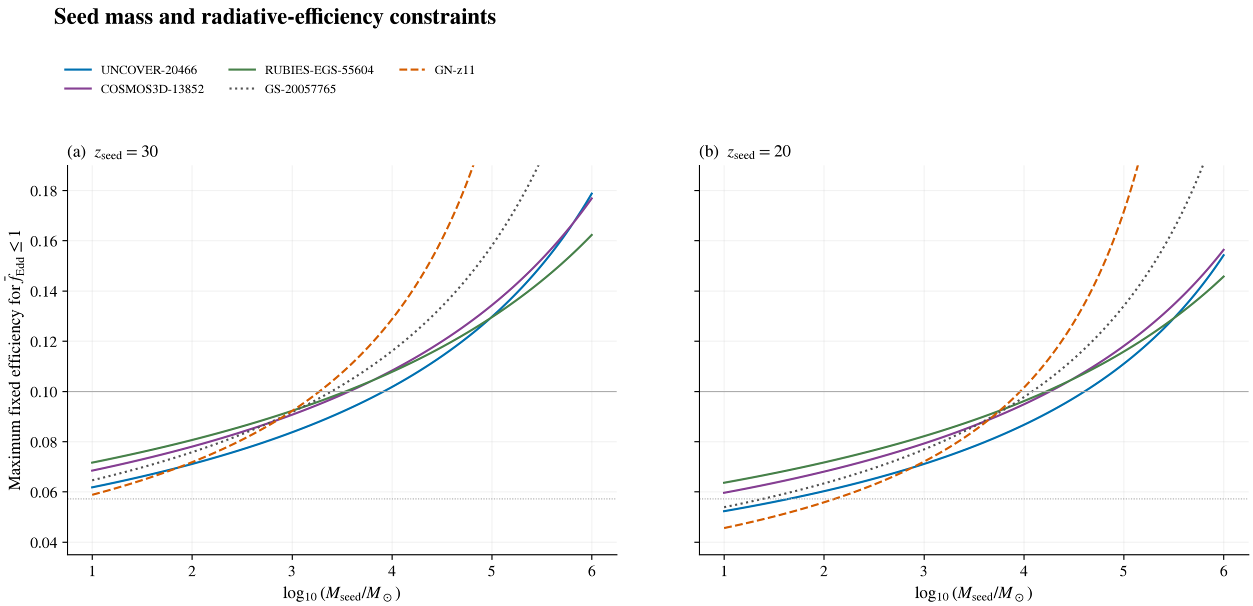

Figure 3: Seed-mass–efficiency boundaries for Eddington-limited average growth at two starting redshifts. Each curve uses an adopted point mass and no merger boost. Below a curve the required average is at most unity. Horizontal guides mark $\epsilon = 0.1$ (solid grey) and $1 - \sqrt{8/9}$ (dotted grey). GN-z11 (dashed) is exploratory; the other objects are primary. These are fixed-efficiency comparisons without probability weights.

Table 5: Controlled comparison for four named literature matches. Columns give required average $f_{\rm Edd}$; the first two share $z_{\rm seed} = 25$. The external inputs and source locators are released with the analysis.

Object	Earlier inputs, 25	Adopted inputs, 25	Adopted inputs, 30
GN-z11	1.577	1.578	1.437
A2744-QSO1	0.974	0.974	0.929
UNCOVER-20466	1.540	1.548	1.452
CEERS-00717	1.074	1.076	1.027

the reference history. We retain the values from arXiv:2508.21748v2; the reference lists the subsequent journal publication. The latter estimate depends on lens reconstruction and inclination; its inclination-free lower limit is not sampled symmetrically. The point requirement rises from 0.929 to 1.000. With a symmetric normal approximation to the quoted log-mass errors, the exact 16th–84th percentile intervals are 0.895–0.964 and 0.947–1.052, respectively. The direct estimate lies just below unity at the point value, but its interval crosses the threshold; the conditional exceedance probability is 0.497. Rounding the point requirement to 1.000 does not indicate exact equality. The dynamical estimate moves the object from below the reference threshold to an interval spanning it. This independent mass check leaves frozen membership unchanged and does not calibrate the full virial sample.

5 Discussion

5.1 What the high-pressure tail establishes

The objects above the reference threshold lie at $z = 6.6214$–10.603; the lower-redshift majority supplies comparison objects. Their growth requirements constrain combinations of seed mass, starting time, effi-

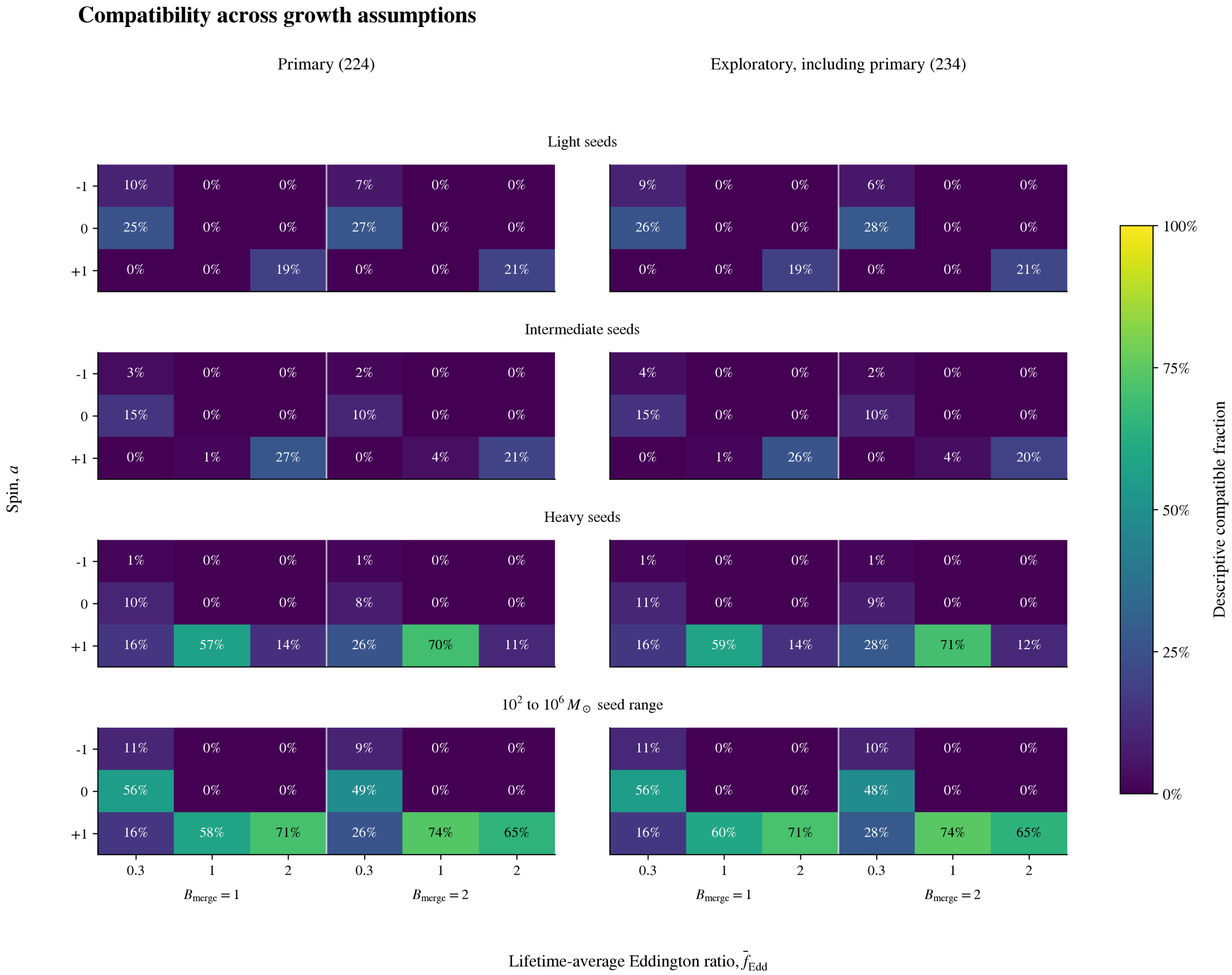

Figure 4: Constant-state compatibility fractions in the primary sample (left) and the inclusive exploratory sample (right). These samples are nested. Each column specifies merger multiplier B and fixed luminosity ratio f; rows specify ideal spin-dependent efficiency, with the supercritical prescription in Appendix C. Seed intervals overlap and carry no probability weights. The broad mass interval has no formation-channel interpretation.

ciency and accretion, rather than selecting a unique formation history.

At fixed seed, growth interval and merger factor, required f_{Edd} scales as $\epsilon/(1-\epsilon)$. Lowering ϵ from 0.1 to 0.05719 multiplies all requirements by 0.54594: the largest becomes 0.793. The reference tail therefore does not establish a model-independent need for super-Eddington accretion or measure black-hole spins. Heavier or earlier seeds and merger growth offer further degeneracies.

Table 6 links the demanding targets to tests of their mass assumptions. Independent masses and constraints on scattering, outflows, obscuration and lensing can distinguish these interpretations. The follow-up ordering is heuristic, not a calibrated estimate of information gain; a high quoted-error probability alone does not validate an estimator.

5.2 Estimator credibility and observational tests

The -0.5 dex test does not bracket every estimator limitation. UNCOVER-20466's demagnified Hβ mass depends on lensing and broad-line interpretation (J. E. Greene et al. 2024); RUBIES-EGS-55604's dust-corrected Hβ mass has a separately recorded 0.5 dex calibration scale (D. D. Kocevski et al. 2025).

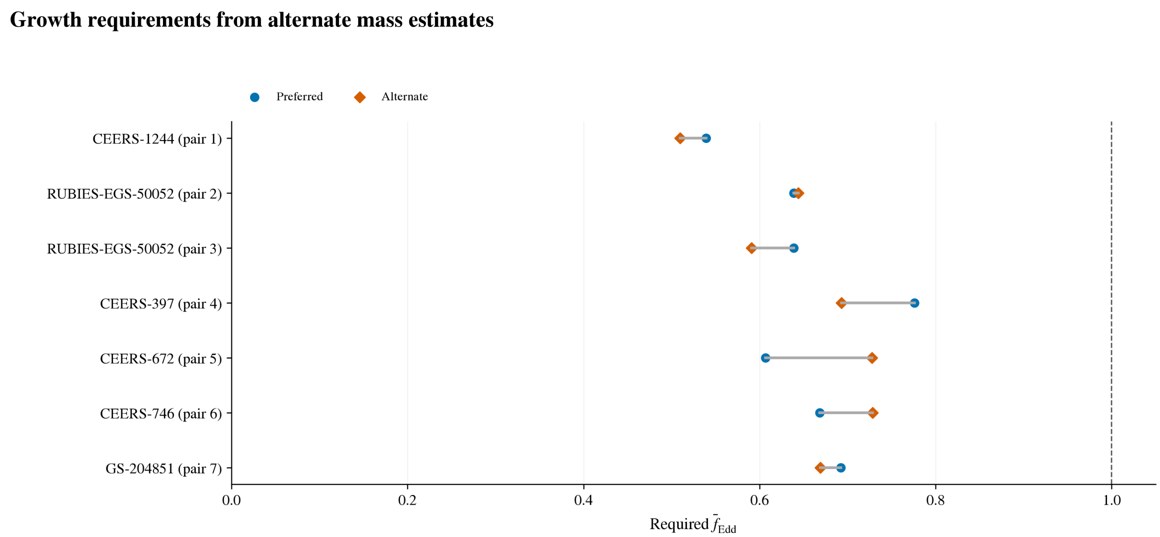

Figure 5: Preferred (circles) and alternate (diamonds) point requirements for seven measurement pairs across six primary objects. Pair numbers follow the released alternate-measurement table, which supplies stable measurement IDs; RUBIES-EGS-50052 (CEERS-2782) has two alternatives. All pairs remain below the reference threshold.

For COSMOS3D-13852, scattering could inflate the Hβ mass by 1–2 dex (X. Lin et al. 2026). Those downward offsets give point requirements of 1.115 and 0.916: illustrative source-motivated tests, not inferred corrections or probability distributions.

J0910_2028_12910 has a 0.01 dex quoted error and a requirement of 1.053, giving $P > 0.999$, yet its separately recorded 0.5 dex calibration scale (C. J. Baccus & X. Xu 2026) allows it below unity in the negative-offset test. In contrast, ZS7's quoted 0.4 dex already includes calibration scatter (H. Übler et al. 2024). Quoted-error probabilities therefore do not rank equivalent physical uncertainties.

5.3 Contrasting object-level constraints

UNCOVER-20466 and GS-20057765 have reference requirements of 1.452 and 1.355, with 16th percentiles above unity. Their point requirements remain above unity for $10^3 M_\odot$ seeds (1.217 and 1.101), but fall to 0.746 and 0.592 for $10^5 M_\odot$ seeds. At $\epsilon = 0.05719$, light seeds require 0.793 and 0.740. For Eddington-limited active phases separated by zero-accretion phases, these imply active fractions of about 79 and 74 per cent of the available time: sub-unity averages can still demand sustained fuelling.

GN-z11's reference requirement is 1.437. Seeding at $z = 20$ raises it to 1.880, above UNCOVER-20466's 1.733, reversing their order. An independent, method-comparable mass would test this timing constraint: GN-z11 is excluded from the primary sample by its UV estimator, not its evidence classification.

5.4 Limitations

The heterogeneous source selections lack a common completeness model and cannot support pooled demographics. Virial calibration and geometry errors are incompletely reported; coherent mass offsets do not model source-specific biases, correlated errors or selection-dependent scatter.

Two source-supported duplicate pairs are reconciled; three groups remain open (Appendix A.2). Excluding all six affected records preserves the threshold results. Readmission requires source-backed spectral, mass, imaging or aperture reconciliation; unique-object and host counts remain provisional.

Table 6: Source-linked interpretation of the complete primary threshold set. Caveats summarize the source records cited in Table 1; proposed observations are our assessment of the measurement that would most directly test the stated limitation, not scheduled or guaranteed feasible programmes.

Object	Mass-estimator caveat	Proposed observation
UNCOVER-20466	Hbeta; demagnified mass; lens model and BLR assumptions remain.	Independent lens model and broad-line profile constraints.
GS-20057765	Hbeta; quoted errors; extinction and calibration not fully quantified.	Deeper line profiles and an extinction constraint.
COSMOS3D-13852	Hbeta; source warns of possible 1–2 dex scattering bias.	Separate scattering wings from virial broadening.
RUBIES-EGS-55604	Hbeta; dust-corrected mass; separate 0.5 dex calibration scale.	Test dust correction and broad-line kinematics.
CEERS-1019	Hbeta; broad component has comparatively low significance.	Confirm the broad component with deeper spectroscopy.
GS-20030333	Hbeta; broad quoted errors; extinction/calibration caveats.	Improve line profile and extinction constraints.
GS-164055	Hbeta; reported interval straddles the reference threshold.	Reduce mass uncertainty with deeper spectroscopy.
GN-38509	Halpha; separate 0.3 dex calibration scale.	Independent estimator or calibration constraint.
J0910_2028_12910	Halpha; 0.01 dex formal error excludes 0.5 dex mass scale.	Independent mass calibration; smaller fit errors alone are insufficient.
CEERS-00717	Halpha; local calibration; no numeric systematic supplied.	Cross-check the virial estimator and line decomposition.
ZS7	Hbeta; quoted 0.4 dex already includes calibration scatter.	Spatially separate nuclear kinematics and interacting components.
GN-4685	Hbeta; reported interval straddles the reference threshold.	Improve line profile and extinction constraints.

Equation 2 compresses accretion, feedback, mergers and radiative changes into effective parameters, not reconstructed histories. Reported-error probabilities condition on fixed growth assumptions; marginalization would require justified parameter distributions and correlations. Neither these probabilities nor scenario fractions measure seed origins. The age relation neglects radiation, so the supplementary $z_{\rm seed} = 3400$ plot is an extrapolation without a physical radiation-era growth interpretation.

Independent source checks cover all admitted mass/error fields, 350 central redshifts and 323 available coordinate pairs; 27 rows lack coordinates. Redshift errors, full posteriors and auxiliary observables are not exhaustively validated. Completeness refers to the declared sources through 3 September 2026, not all JWST discoveries. The external A2744-QSO1 test (Section 4.6) leaves this frozen membership intact.

6 Conclusions

The atlas provides a reproducible, source-linked comparison of early black-hole growth requirements. Its principal results are:

- The conservative 224-object primary sample contains 12 reference point requirements above unity and six with conditional $P \geq 0.95$ under quoted errors.
- UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604 retain the probability threshold after a -0.5 dex mass shift. Their distinct lensing, scattering, and calibration caveats identify different observational tests.

- Seed formation time changes both threshold membership and the most demanding object; lowering the fixed efficiency to 0.05719 removes all point requirements above unity. These are constraints on combinations of assumptions.
- The external A2744-QSO1 dynamical estimate increases its reference requirement from 0.929 to approximately unity, with an error interval spanning the threshold. Independent mass information can change the growth interpretation even without adding a new object.

The threshold results survive exclusion of the known ambiguous identities; the full catalogue's unique-object census remains provisional. The useful outcome is a set of named observational tests with explicit assumptions, not a universal ordering or a measurement of formation-channel frequencies.

Appendix A Source-family inventory

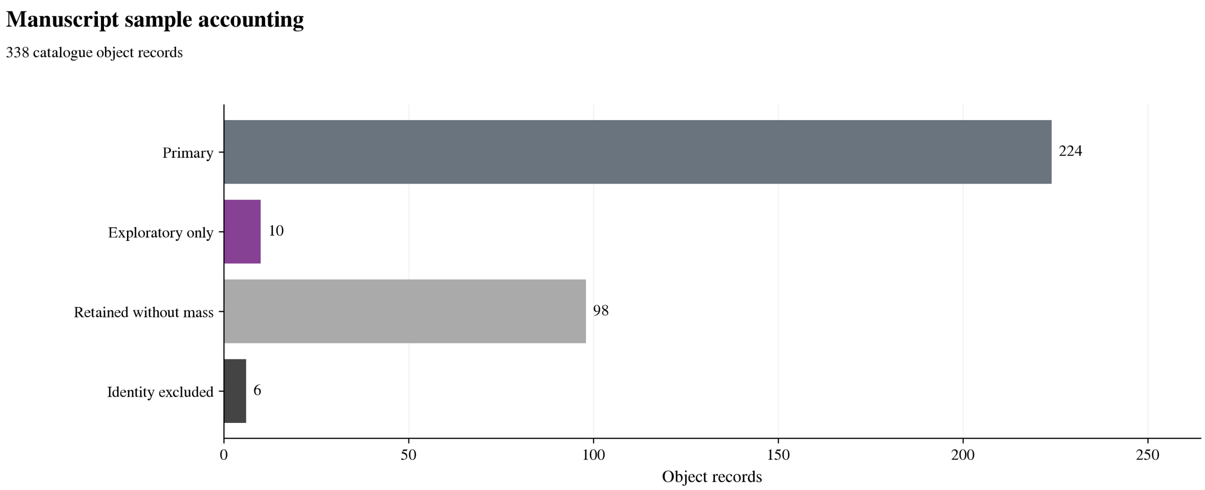

Figure 6: Sample accounting for 338 catalogue object records. Mutually exclusive categories comprise 224 primary objects, ten additional exploratory objects, 98 retained without eligible masses and six identity exclusions. The inclusive exploratory numerical sample contains 234 objects.

Table 7: v3 scale and analysis subsets.

Product or subset	Rows or objects
Measurements / provisional objects / hosts	350 / 338 / 337
Growth-eligible measurements / objects	244 / 237
Method-eligible primary measurements / objects	234 / 227
Manuscript primary / exploratory objects	224 / 234
Identity-excluded object records	6
Catalogue-only object records	101

The full catalogue contains 247 broad-line AGN, 49 narrow-line candidates, 30 photometric candidates, ten high-ionization candidates, and two X-ray candidates. Preferred-row evidence labels comprise 214 secure, 41 probable, 82 candidate, and one disputed classification. These full-catalogue counts precede the manuscript exclusions.

Table 8 gives every admitted source family. Counts distinguish measurements from preferred objects and avoid attributing one shared physical object independently to multiple sources. Auxiliary coordinates are attributed in the registry to THRILS (T. A. Hutchison et al. 2025), UHZ1 spectroscopy (A. D. Goulding et al. 2023), and the versioned JADES DR3 GOODS-S prism catalogue. The machine-readable companion retains full selection-channel and caveat tags alongside source URLs. “Context” and “Limit” denote values retained outside numerical growth inference.

Table 8: Source-family inventory. Counts are admitted measurements / eligible mass measurements / preferred primary objects after identity exclusion (M/G/P). The columns sum to 350/244/224; G includes alternatives, whereas P counts each selected object once. "Balmer" denotes single-epoch virial masses.

Source	M/G/P	Mass	Selection channel; principal caveat
C. J. Baccus & X. Xu (2026)	49/49/48	Balmer	NIRSpec broad lines; Frozen preprint masses; calibration
Ó. A. Chávez Ortiz et al. (2026)	1/0/0	Context	UV/optical diagnostics; Model-dependent mass
J. Chisholm et al. (2024)	1/0/0	None	High-ionization lines; No canonical mass
K. Davis et al. (2026) / THRILS	6/6/6	Balmer	EELG-selected spectroscopy; Virial calibration
Q. Fei et al. (2026) / GLIMPSE	10/10/8	Balmer	Lensed broad lines; Lensing corrections
J. E. Greene et al. (2024) / UNCOVER	9/9/9	Balmer	Lensed broad lines; Lens model; BLR interpretation
Y. Harikane et al. (2023)	10/10/5	Balmer	NIRSpec broad lines; Local virial calibration
I. Juodžbalis et al. (2026a) / JADES	23/23/21	Balmer	Broad Balmer lines; Extinction; calibration
M. Killi et al. (2024)	1/1/1	Balmer	Broad Halpha; Linear mass-error conversion
D. D. Kocevski et al. (2025) / LRDs	6/6/6	Balmer	Broad Balmer lines; Dust correction; calibration
R. L. Larson et al. (2023) / CEERS-1019	1/1/1	Balmer	Broad Hbeta; Broad-component significance
X. Lin et al. (2024) / ASPIRE	16/16/16	Balmer	Broad Halpha; Virial calibration
X. Lin et al. (2026) / COSMOS-3D	13/13/13	Balmer	NIRCam grism broad lines; Possible scattering bias
J. Lyu et al. (2024) / SMILES	19/0/0	None	MIRI SED selection; SED-dependent AGN identification
R. Maiolino et al. (2024) / GN-z11	1/1/0	UV	UV lines; UV estimator outside primary
S. Mascia et al. (2026)	8/0/0	None	Compact blue broad emitters; No canonical mass
J. Matthee et al. (2024) / EIGER-FRESCO	20/20/19	Balmer	Broad Halpha; Dust; virial calibration
G. Mazzolari et al. (2025)	25/0/0	None	Narrow-line diagnostics; Star-formation overlap
G. C. K. Leung et al. (2026) / MEOW	12/0/0	None	MIRI SED selection; SED-dependent AGN fraction
R. P. Naidu et al. (2026) / MoM-BH*-1	1/0/0	Context	Broad Hbeta / SED; Scattering; stellar alternative
L. Napolitano et al. (2025a) / GHZ9	1/0/0	Context	UV lines / X-ray evidence; Mass assumes Eddington rate
L. Napolitano et al. (2025b) / GHZ4,7	2/0/0	None	Narrow / high-ionization lines; Tentative diagnostics
W. Ren et al. (2025) / ALPINE-CRISTAL	7/7/1	Balmer	Host-selected broad Halpha; Conditional BLR interpretations
J. Scholtz et al. (2025) / JADES	21/0/0	None	Narrow / high-ionization lines; Tentative classifications
D. D. Kocevski et al. (2026) / Skyfire (CEERS)	22/22/22	Balmer	Broad Halpha; Virial calibration
M. Tang et al. (2025)	3/0/0	None	High-ionization lines; Ionizing-source ambiguity
A. J. Taylor et al. (2025) / CEERS-RUBIES	37/37/35	Balmer	Broad Halpha; Slit loss; dust; calibration
H. Treiber et al. (2025) / UNCOVER	2/0/0	None	UV-line diagnostics; Stellar alternatives

Table 8 continued

Source	M/G/P	Mass	Selection channel; principal caveat
H. Übler et al. (2024) / ZS7	1/1/1	Balmer	Offset broad Hbeta; Error includes calibration scatter
Á. Bogdán et al. (2024); F. Zou et al. (2026) / UHZ1	2/0/0	Context	X-rays / SED; Later detection disputed
Z. Zhang et al. (2026) / LRDs	5/0/0	Limit	Narrow lines / SED; Mass upper limits
M.-Y. Zhuang et al. (2026) / NEXUS	15/12/12	Balmer	NIRCam WFSS broad lines; Twelve masses lack errors

A.1 Source-admission checks

The v3 additions include 15 NEXUS measurements from NIRCam/WFSS (M.-Y. Zhuang et al. 2026), 13 COSMOS-3D NIRCam-grism measurements (X. Lin et al. 2026), and GHZ4/GHZ7 (L. Napolitano et al. 2025b). These contribute 30 measurements and 29 object records. NX10835 lies 0.066 arcsec from an existing Mascia identifier at consistent redshift, so the new mass-bearing measurement becomes preferred for the same object.

The Scholtz et al. source audit provides another cardinality check. The paper reports 42 objects, whereas its source-native table contains 41 rows. Exactly 21 table rows satisfy $z \geq 4$; all are admitted. One is identity-linked to a broad-line object, yielding 20 distinct narrow-line candidates from that source. JADES-NS-GS00099671 is retained with its published host and bolometric observables but without an invented black-hole mass.

A.2 Preferred measurements and identity exclusions

Stable measurement, object and host identifiers retain extraction locations, source versions and uncertainty semantics. Preferred measurements are chosen from source assessments, not inferred growth pressure or publication date alone. The CEERS-2782 group uses RUBIES-EGS-50052 because the source prefers its higher-signal-to-noise spectrum over the slit-loss measurement. NX10835 uses the newly available NEXUS mass; UHZ1 uses the later full-dataset evidence assessment and remains outside numerical growth inference. The measurement–object link table records each choice and its reason.

Before the conservative identity exclusions, there are 244 eligible measurements and 237 preferred objects: 236 Balmer-line and one UV single-epoch virial masses. The method-defined primary subset has 234 measurements and 227 objects. The six excluded records link to seven admitted measurements; the three with eligible masses are GDS_1210_9515, GS-8083 and GS-10013704. All three otherwise pass the primary cuts. The other three excluded records lack eligible masses. Exclusion does not establish duplication or resolve the identities. The open groups are Baccus GDS_1210_9515 versus JADES GS-8083, GS-10013704 versus Scholtz 99671, and Scholtz 16745 versus 208643. The first requires target/spectral and preferred-mass reconciliation; the latter two require imaging and aperture checks. Seven of the nine additional candidate objects have conditional broad-line interpretations. All original measurements and catalogue flags remain available.

Appendix B Catalogue navigation score

The point navigation score is

$$S = \min\left[100, 100 \max\left(C\left(\frac{f_2 - 0.3}{1.2}\right), C\left(\frac{s_{0.3} - 4}{2.8}\right)\right) + 8C\left(\frac{z-6}{4}\right)\right], \tag{4}$$

where $C(x) = \min(1, \max(0, x))$, f_2 is the required $f_{\rm Edd}$ for a $10^2\,M_\odot$ seed, and $s_{0.3}$ is the required $\log_{10}(M_{\rm seed}/M_\odot)$ at $f_{\rm Edd} = 0.3$. This heuristic combines two growth diagnostics and a redshift term; it is not a probability. Ties use decreasing f_2, decreasing redshift, then stable identifier. The manuscript's required-$f_{\rm Edd}$ table is instead ordered directly by f_2.

Appendix C Efficiency prescription for compatibility scenarios

For the illustrative spin-dependent maps, the ideal Kerr thin-disk efficiency is the ISCO binding efficiency (J. M. Bardeen et al. 1972):

$$\epsilon_a = 1 - \sqrt{1 - \frac{2}{3r_{\rm ISCO}}}, \tag{5}$$

$$r_{\rm ISCO} = 3 + Z_2 - \mathrm{sgn}(a)\sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}, \tag{6}$$

$$Z_1 = 1 + (1 - a^2)^{1/3}\left[(1 + a)^{1/3} + (1 - a)^{1/3}\right], \qquad Z_2 = \sqrt{3a^2 + Z_1^2}. \tag{7}$$

The dimensionless spin is $a = cJ/(GM_{\rm BH}^2)$, where J is the signed angular momentum relative to the disk, and $r_{\rm ISCO}$ is in units of $GM_{\rm BH}/c^2$. The ideal efficiencies at $a = -1, 0, +1$ are approximately 0.038, 0.057, and 0.423, respectively, neglecting photon capture and stress at the ISCO. Logarithmic luminosity growth at supercritical inflow is motivated by advection and wind models (J. Poutanen et al. 2007). Our unit-coefficient relation is a phenomenological ansatz, not a fit to those models or a treatment of radial mass loss. Above unity, the maps adopt

$$f_{\rm Edd} = 1 + \ln \dot{m}, \qquad \dot{m} = \frac{\dot{M}_{\rm in}}{L_{\rm Edd}/(\epsilon_a c^2)}, \qquad \epsilon_{\rm eff} = \begin{cases} \epsilon_a, & f_{\rm Edd} \le 1, \\ \epsilon_a f_{\rm Edd} \exp(1 - f_{\rm Edd}), & f_{\rm Edd} > 1. \end{cases} \tag{8}$$

The logarithmic luminosity ansatz applies only on the super-Eddington branch. Each compatibility cell holds both the rate and its effective efficiency constant over the interval. Thus the cell cannot represent a variable history merely by substituting its average luminosity ratio. This is an illustrative parameter-scan prescription, not a relativistic slim-disk or spin-evolution calculation. If efficiency varies with time, the integrated growth depends on $\int[(1-\epsilon(t))/\epsilon(t)]f_{\rm Edd}(t)\,dt$; it cannot in general be represented by separately averaged efficiency and rate.

Figure 4 gives the full scenario grid. The broad 10^2–$10^6 M_\odot$ interval retains the legacy code key `pbh_origin_hypothesis` solely for artifact compatibility; it models no primordial formation process. The descriptive fractions use point masses and the inclusive interval criterion in Section 3.3, without error integration. Appendix D.4 explains the conditional interpretation of this interval for pre-existing primordial black holes.

Appendix D Growth-model derivations and implementation

D.1 Accretion and exponential growth

The growth constraints follow from the relation between accretion luminosity, retained mass, and available cosmic time. For electron-scattering opacity in ionized hydrogen, the Eddington luminosity and luminosity-based Eddington ratio are

$$L_{\rm Edd} = \frac{4\pi G M_{\rm BH} m_p c}{\sigma_T}, \qquad f_{\rm Edd} = \frac{L_{\rm bol}}{L_{\rm Edd}}, \qquad t_{\rm Edd} = \frac{M_{\rm BH}c^2}{L_{\rm Edd}} = \frac{c\sigma_T}{4\pi G m_p} \simeq 0.45\ \text{Gyr}. \tag{9}$$

Here G, c, m_p, and σ_T denote the gravitational constant, speed of light, proton mass, and Thomson cross section. With radiative efficiency $0 < \epsilon < 1$ and rest-mass inflow rate $\dot{M}_{\rm in}$,

$$L_{\rm bol} = \epsilon \dot{M}_{\rm in} c^2, \qquad \dot{M}_{\rm BH} = (1-\epsilon)\dot{M}_{\rm in} = \frac{1-\epsilon}{\epsilon}\frac{f_{\rm Edd}}{t_{\rm Edd}} M_{\rm BH}. \tag{10}$$

The inflow rate is defined at the radiating flow; additional mass loss before that point is not modelled. Dividing Eq. 10 by $M_{\rm BH}$ and integrating at fixed ϵ gives

$$\ln\left(\frac{M_{\rm BH}}{B_{\rm merge} M_{\rm seed}}\right) = \frac{1-\epsilon}{\epsilon}\frac{1}{t_{\rm Edd}} \int_{t(z_{\rm seed})}^{t(z_{\rm obs})} f_{\rm Edd}(t)\, dt = \frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}}. \tag{11}$$

Exponentiating yields Eq. 2; solving for the average rate and applying the non-negative floor yields Eq. 3. For a constant positive $f_{\rm Edd}$, the e-folding time is

$$t_{\rm efold} = \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{f_{\rm Edd}}. \tag{12}$$

It is 0.05 Gyr at $\epsilon = 0.1$ and $f_{\rm Edd} = 1$.

D.2 Merger multiplier and inverse seed mass

The implementation evaluates Eq. 2 in log mass:

$$\log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm seed}}{M_\odot}\right) + \log_{10} B_{\rm merge} + \frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}\ln 10}. \tag{13}$$

We evaluate $B_{\rm merge} = 1$ (no boost) and 2 (a factor-of-two mass boost). At fixed accretion history,

$$\frac{M_{\rm BH}(B_{\rm merge}=2)}{M_{\rm BH}(B_{\rm merge}=1)} = 2, \qquad \Delta \log_{10} M_{\rm BH} = \log_{10} 2 \simeq 0.3010. \tag{14}$$

Solving the same relation for seed mass gives

$$\log_{10}\left(\frac{M_{\rm seed,req}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) - \log_{10} B_{\rm merge} - \frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}\ln 10}. \tag{15}$$

Here $M_{\rm seed,req}$ is the seed mass required for a specified average rate, efficiency, growth interval and merger multiplier.

D.3 Cosmic time

Cosmic ages use the flat matter-plus-Lambda relation

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \,\mathrm{asinh}\left[\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}(1+z)^{-3/2}\right], \tag{16}$$

with $H_0 = 67.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$. The Hubble constant is converted to inverse time; ages and growth intervals are in Gyr. Radiation is neglected. In particular, the supplementary $z_{\rm seed} = 3400$ comparison extrapolates this age relation and is not a radiation-era seed-formation calculation.

D.4 Earlier growth onset and primordial seed interpretations

The growth equations depend on starting mass and subsequent accretion, rather than the seed's formation channel. A primordial black hole (PBH) already present at $z = 30$ can therefore be represented by its effective mass at that epoch if it lies within the broad 10^2–$10^6 M_\odot$ interval and follows the adopted subsequent growth prescription. This mass need not equal its primordial formation mass. The interval accommodates such conditional PBH interpretations without identifying a primordial origin or covering all PBH seed masses. In P. Dayal (2024), $z \simeq 3400$ marks matter–radiation equality and the start of the modeled evolution of pre-existing PBHs, not their formation epoch.

Moving the start of the same constant-rate, constant-efficiency history from $z = 30$ to 3400 adds $\delta t = t(30) - t(3400) = 0.09990$ Gyr in our matter-plus-Lambda extrapolation. At fixed seed mass and merger multiplier, the resulting shift is

$$\Delta \log_{10} M_{\rm BH} = \frac{1-\epsilon}{\epsilon} \frac{\overline{f_{\rm Edd}}\,\delta t}{t_{\rm Edd} \ln 10}. \tag{17}$$

For $\overline{f_{\rm Edd}} = 1$, this is 0.868 dex (a factor of 7.37) at $\epsilon = 0.1$, or 1.589 dex (a factor of 38.85) at the ideal non-spinning thin-disc efficiency. Although the tracks retain their shape, these mass shifts are substantial (Figure 7). Conversely, at fixed observed mass and seed mass requiring positive accretion, the ratio of required average rates is $[t(z_{\rm obs}) - t(30)]/[t(z_{\rm obs}) - t(3400)]$; the earlier start lowers the requirement by 6.5, 13.1 and 22.9 per cent at $z_{\rm obs} = 4$, 7 and 10.603, respectively. Similar-looking tracks on a wide logarithmic plot therefore do not establish insensitivity to the starting epoch.

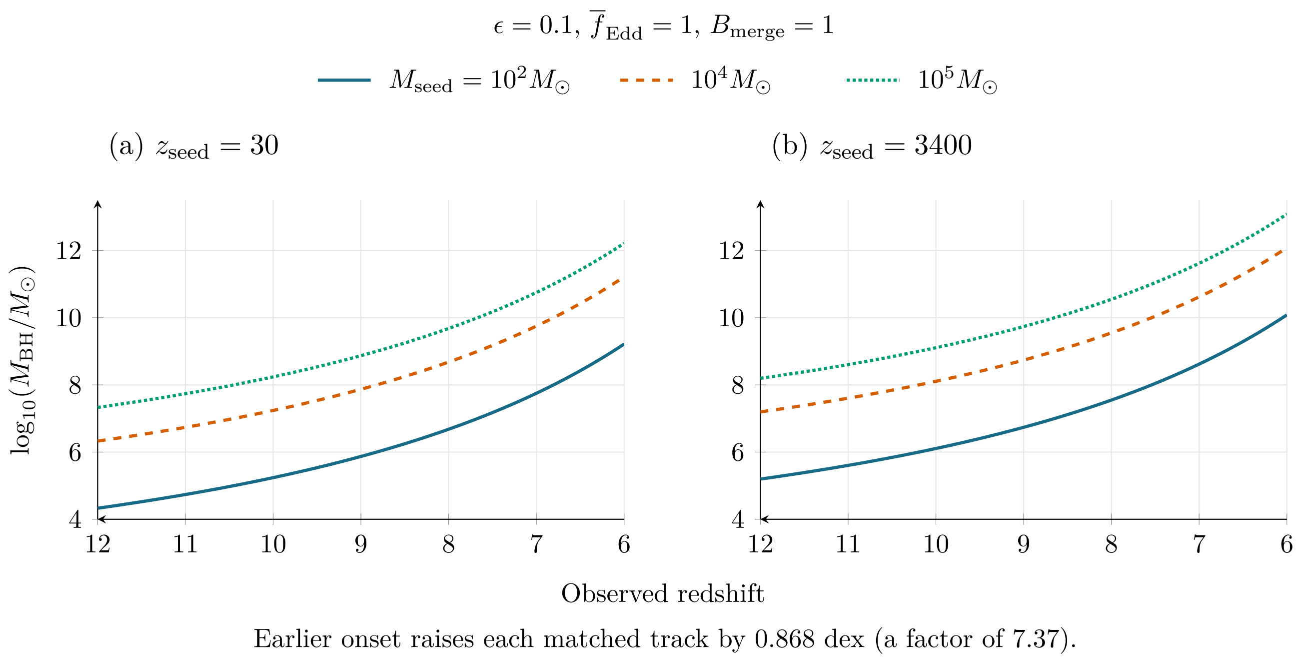

Figure 7: Matched growth tracks with $z_{\rm seed} = 30$ (a) and $z_{\rm seed} = 3400$ (b). Here $z_{\rm seed}$ denotes the epoch at which the initial mass is assigned and accretion begins, not necessarily the formation epoch of a PBH. Colours and line styles identify seeds of 10^2, 10^4 and $10^5 M_\odot$; all tracks use $\epsilon = 0.1$, $\overline{f_{\rm Edd}} = 1$ and no merger boost. Both panels use identical axis limits. Moving the start earlier adds 99.90 Myr and raises every matched track by 0.868 dex (a factor of 7.37), without changing its shape. The displayed range $6 \le z \le 12$ focuses on early growth. These are controlled analytic comparisons, not fitted histories. The $z = 3400$ curves extrapolate the matter-plus-Lambda age relation and assume uninterrupted accretion; they do not model PBH formation or establish that such early gas accretion occurs.

These comparisons isolate the extra time allowed by the analytic prescription. They do not establish continuous gas accretion from equality: radiation, gas supply, halo assembly and feedback are not modeled here. PBH compatibility must consequently be stated in terms of the effective mass and subsequent growth assumptions, rather than inferred from visual similarity of the tracks.

D.5 Two-state duty cycle

For a two-state history with the same efficiency in both states,

$$\overline{f_{\rm Edd}} = D f_{\rm burst} + (1-D) f_{\rm quiet}, \qquad D_{\rm req} = \frac{\overline{f_{\rm Edd}}_{\rm req} - f_{\rm quiet}}{f_{\rm burst} - f_{\rm quiet}}. \tag{18}$$

Here D is the active-state time fraction, $f_{\rm burst} > f_{\rm quiet} \geq 0$, and the diagnostic requires $\overline{f_{\rm Edd}}_{\rm req} \geq f_{\rm quiet}$. Values $D_{\rm req} > 1$ identify an infeasible specified two-state history and are retained. An average growth requirement is not an instantaneous Eddington ratio or a uniquely measured duty cycle.

D.6 Sampling, mass conversions and track selection

Draws use seed 20260808 and a stable identifier-based random stream. For 10,000 draws, the binomial Monte Carlo standard error of a probability is at most 0.005 (approximately 0.0022 at probability 0.95). This is numerical sampling error, separate from astrophysical uncertainty. The twelve mass-bearing NEXUS objects have no published mass errors: repeated point values produce zero-width stored intervals, while probabilities and uncertainty ranks remain unavailable. Source errors are retained as quoted and need not be purely statistical. For ZS7, the published 0.4-dex error already includes virial-calibration scatter. Separately reported method systematics are stored and, where possible, mapped to sensitivity envelopes without assuming independent Gaussian contributions.

The mass-scale tests translate each object's original draws by the fixed offset; no scatter is added in quadrature and no marginalization over the offset is performed. The final published Killi mass $8^{+0.5}_{-0.4} \times 10^8 M_\odot$ is converted by transforming both linear bounds, giving $\log_{10}(M/M_\odot) = 8.90309^{+0.02633}_{-0.02228}$.

For Figure 1, candidate average rates span 0.1–2.0 in steps of 0.1. For each seed mass and fixed efficiency, we retain the two rates with the most primary objects within 0.5 dex of either the $B = 1$ or $B = 2$ prediction, requiring at least five such objects. Ties favour smaller median absolute residuals, then smaller rates. Objects at $z \simeq 4$–6 dominate this display selection; the tolerance is not an uncertainty interval. Baseline rankings, duty-cycle diagnostics and seed-redshift–mass maps retain $\epsilon = 0.1$. The heuristic catalogue score in Appendix B is separate from the required-rate ranking used for scientific conclusions.

Appendix E Data and code availability

The catalogue, source registries, code, and manuscript analysis are available at `https://github.com/nniickels/highz-accretion-atlas`. The frozen v3 catalogue baseline is identified by Git commit a40a0d2; version manifests under `releases/` record the dataset artifact hashes. The revised manuscript's source-linked comparisons and generated tables are in `paper/analysis/`, with explicit external inputs in `paper/review_inputs.json`. A permanent archival DOI has not yet been assigned; the repository and Git baseline provide the current access and version identification.

Five ordered notebooks reproduce the catalogue, science products, figures, and atlas using pinned dependencies. Reproduction checks compare regenerated numerical products, figures, and manuscript table fragments with an independent stored baseline, then verify their internal consistency. Independent cosmic-age quadrature and growth integration supplement the regression checks. The source-family inventory in Appendix A links measurements, methods, selection channels, and major caveats. Full identifiers, source-native observables, alternative measurements, and per-object maps remain available.

Source agreement does not establish estimator validity or astrophysical uniqueness. The manuscript exclusion check verifies that all open identity groups are withheld; the separate full-catalogue identity gate remains open for the three groups listed in Section 5.4.

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